Proposal 1: Is Generative A.I. the Future of Education? The Risks Involved

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The rapid emergence of **generative artificial intelligence** (GenAI)—a subset of AI that "uses deep learning techniques to create new content such as images, music, animation, 3D models, and text" (University of Illinois Urbana-Champaign, 2024)—has introduced a new wave of transformative tools into the school setting, with students reshaping what it means to learn, practice, and interact with educational content. From chatbot conversations in Gemini to personalized instruction in **educational technology (EdTech)** platforms like Duolingo and Khan Academy, GenAI has become substantial in providing students with instant feedback, step-by-step guidance, and access to a wide range of subjects beyond the traditional classroom.

On one hand, most EdTech platform companion tools that use GenAI are based on GPT (Generative Pre-trained Transformer) models, which use "self-supervised learning to pre-train massive amounts of text data, enabling it to generate high-quality [responses]" (Yenduri et al., 2024). In other words, these models are trained on large datasets that allow them to contextualize and answer user prompts. Despite delivering responses in seconds, the accuracy of GPT-generated output is not guaranteed, and errors in explanations are common. While the speed and availability of these tools offer students support at any time of day, they also carry the risk of undermining information recall and long-term retention.

On the other hand, chatbots continue to be the most popular choice amongst most students. According to recent Pew Research Center surveys (2025), **about 26% of U.S. teens say they have used ChatGPT to help with their schoolwork—doubling the share reported in 2023—and roughly 3 in 10 teens report using AI chatbots daily.** In short, individualized support that can adapt to their pace, style, and needs has become the selling point of these technologies for middle and high school students—particularly those in grades 8 through 12—as reflected in recent usage trends.

Nonetheless, the other side of the coin points to concerning limitations posing a risk to students' **learning retention**: a lack of contextual awareness, hallucinations, occasional inaccuracies, and a focus on providing answers rather than fostering deep understanding. Misperceptions about the reliability of AI, coupled with an incomplete oversight and guidance from educational institutions, can lead students to rely too heavily on these tools and potentially undermine their critical thinking and problem-solving skills.

This analysis examines the risks of GenAI across K–12 classrooms, with particular attention to grades 5–12, highlighting factors that affect student learning retention and the gaps between technological capability and educational outcomes.

The Importance of Learning Retention

Learning retention consists of recalling information about a given subject and is typically assessed in school through comprehensive exams (e.g., end-of-unit quizzes). Its importance lies in the foundational skills students build while contemplating the next semester, the next school year, and situations beyond these formative years (MIT Teaching + Learning Lab, 2026).

Three learning strategies are suggested to report learning retention: (1) leveraging a student's prior knowledge, (2) engaging students in the retrieval of previously learned information, and (3) promoting deep understanding through self-explanation (MIT Teaching + Learning Lab, 2026). The following paragraphs will discuss how GenAI tools pose risks to these strategies and what consequences could lie ahead.

The Risks

Limited Reflection of Student Thinking in AI Feedback

One of the main issues with GenAI tools, even those provided by EdTech companies, is the inability to accurately reflect students' thinking in their responses. For instance, in an article written by high school teacher Dan Meyer (2024), he asserts that Khan Academy feeds student thinking into the platform's GenAI tool, Khanmigo, "only sometimes and only sometimes accurately." Meyer also highlights how Khanmigo usually begins user interaction with the same type of prompt, questions students about basic steps even after demonstrating understanding, and dismisses what the students did correctly.

More broadly, research shows that "several critical issues which hinder the practicality of GenAI technologies for feedback provision," including reliability (e.g., hallucinations), accuracy for "authentic learning environments," and acceptance amongst students (Li et al., 2025).

These problems can lead to students' frustration or disengagement with the study material as their correct reasoning is overlooked, reducing confidence and willingness to solve challenging problems later. In college, being unable to use "key cognitive strategies" such as analysis and reasoning can lead to a student underperforming in their classes (Conley, 2007).

Increasing Dependency on GenAI

As noted by Monash University researcher Lixiang Yan (2024), overreliance on GenAI "for learning content creation […] without validation could introduce inaccuracies, misleading both educators and students." For instance, if students typically use a chatbot to summarize texts or generate responses for assignments, their independent analysis skills may diminish over time. Understanding that platforms like Khan Academy and Duolingo leverage large language models (LLMs) such as those developed by OpenAI—including models that power ChatGPT—we can expect similar intrinsic flaws to manifest in their respective EdTech tools.

Regarding this ongoing concern, a recent study published in Proceedings of the National Academy of Sciences (PNAS) found that while GenAI tools can "substantially improve human performance when access [to them] is available, they can also degrade human learning (particularly when appropriate safeguards are absent), which may have a long-term impact on human performance" (Bastani et al., 2025).

This dynamic relates to what recent literature describes as an "**agency gap,**" the extent to which GenAI literacy predicts student writing performance in contexts that require self-initiation and regulation (Jin et al., 2025).

Superficial Learning Over Deep Understanding

As noted in a recent report published by Microsoft Research, "GenAI's productivity improvements in industry might not be desirable in education settings," where the focus is centered more on fostering critical thinking and social learning than on finishing tasks faster (Walker & Vorvoreanu, 2025). Inside the classroom setting, general-use GenAI tools supply quick answers to complex questions, but the lack of teacher intervention doesn't help students avoid receiving inaccurate responses, biases, or wrong feedback (University of Cincinnati Libraries, 2025). Not only are students tempted to get satisfying results quickly, but they also conform to the arbitrary perception of GenAI being always correct, as previously pointed out.

Research suggests that students often prioritize efficiency and convenience when using GenAI, even when they recognize that such use may not benefit their long-term learning:

> "We found that students have misconceptions about what GenAI is, even when they use it frequently, and largely see it as a tool for making things more efficient. They are also conflicted by its use as they see its benefits but realize it might impede learning. Ideally, they want GenAI to be a coach or tutor that can personalize the learning experience for them."
>
> (Johri et al., 2024)

Moreover, when these flaws go unrecognized, they can foster a dangerous misconception that ultimately leads to academic underperformance. For example, the PNAS study identified a "mismatch between perceived and actual learning," most noticeably in students learning with access to a base GPT model without safeguards, who performed 17% worse on average compared to the control group (Bastani et al., 2025). This misconception can extend beyond students to educators who integrate these tools without fully accounting for their limitations.

Finally, in EdTech platforms like Duolingo, the use of chatbots comes with features that allow users to skip through automated feedback, where—unlike in-person tutoring or traditional self-studying—students continue to separate themselves from a problem-solving process that is still "unclear" to them. This can lead to knowledge gaps throughout the school year and increase cognitive bias. Maintaining these problems beyond high school may hinder job expectations and not prepare students enough to tackle harder courses in college, where GenAI typically cannot answer questions as accurately.

Academic Integrity and the Misuse of GenAI

Along with learning retention, GenAI introduces significant challenges related to **academic integrity**, particularly when students use these tools to complete assignments with minimal independent effort. Academic integrity is defined—in this case—as acting with "honesty, trust, fairness, respect, responsibility, and courage" (International Center for Academic Integrity), values compromised as GenAI tools become more integrated into daily schoolwork.

On this same note, according to a recent UNESCO report, "GenAI might allow students to pass off text that they did not write as their own work, a [novel] type of 'plagiarism'" (Miao & Holmes, 2023). The same report notes that, despite efforts such as AI-generated content watermarks and detection tools, there is still little evidence that these measures or tools are effective. As a result, the boundary between assistance and substitution becomes increasingly unclear.

As students rely on GenAI to generate responses, summarize content, or complete assignments, they may shift away from producing original work toward curating AI-generated outputs, blurring authorship and accountability.

Over time, this not only undermines the principles of academic integrity but also reinforces patterns of overreliance and superficial engagement, where the focus shifts from learning and understanding to efficiency and task completion.

Addressing the Problem

Some solutions currently being considered include active user engagement and scaffolding learning—where guidance and support are gradually removed "as students learn and become more competent" (University at Buffalo, 2024). While these solutions align with established learning science principles, they remain insufficient to fully mitigate the risks to learning retention previously outlined. Thus, a more targeted set of measures can be proposed to improve the outlook of GenAI-powered tools in education.

Human-AI Collaboration

Creating a hybrid model in which educators serve as middlemen in AI-student interactions can help address several of the identified risks. This includes both teachers using GenAI tools to enhance the quality of their feedback and directly overseeing how students interact with these systems.

For example, recent research shows that "GenAI-enabled natural-language insights significantly improved educators' feedback quality compared to educators without AI support" (Li et al., 2025). Rather than replacing teachers, these systems can become assistive tools to help interpret student responses more efficiently while still maintaining human judgment.

Likewise, a more direct teacher oversight approach allows for the correction of inaccurate or misleading AI responses, while also enabling educators to evaluate whether a student's reasoning is being properly reflected in the feedback they receive. We describe this approach as **teacher-in-the-loop** AI systems—middleware frameworks that allow teachers to monitor student interactions, influence or constrain AI outputs, and detect behaviors like copy-pasting. Complementary implementations, such as teaching-facing dashboards, can further provide real-time insights into student engagement and reasoning patterns, allowing educators to intervene when necessary and ensure that AI-supported learning remains aligned with instructional goals.

Well-Defined Educational Policies

Instructional guidance is necessary to regulate how GenAI tools are used in educational settings, beyond the classroom-level interventions. While EdTech companies are increasingly transparent about the limitations of their systems, many school districts still lack clearly defined policies for responsible AI use. This creates ambiguity for both students and educators, particularly in distinguishing between acceptable assistance and academic misconduct. According to UNESCO, GenAI systems may enable students to present AI-generated content as their own, raising concerns about authorship and accountability (Miao & Holmes, 2023).

To address this, local school districts have begun implementing comprehensive guidelines that define appropriate use cases for GenAI, establish boundaries for academic integrity, and promote AI literacy among students and staff. While these efforts represent meaningful progress, policies should further emphasize verification practices—encouraging students to corroborate AI-generated information with trusted sources—and require transparency when AI tools are used in assignments. Additionally, these guidelines should be made readily accessible to parents and families to ensure that responsible usage extends beyond the classroom.

For instance, one of the most established school districts in the state of Maryland, Montgomery County Public Schools, manages an *Online Digital Tools Database* that lists approval status (approved, use responsibly, or prohibited), their privacy policy and terms of service, as well as usage recommendations. Building on this model, districts should introduce specific guidelines for standalone chatbot platforms (e.g., ChatGPT, Gemini), along with clear disclaimers regarding the use of embedded AI features within educational applications such as Duolingo or Khan Academy. Finally, this process can be strengthened through structured input from parents and guardians—such as suggesting new platforms for review—while assigning district officials the responsibility of maintaining and regularly updating these resources. A more informed school ecosystem can deter potential misuse of these tools.

EdTech-School Collaboration

Another key approach involves closer collaboration between EdTech companies and school districts to ensure that GenAI tools align with curricular goals and educational standards. Rather than deploying generalized AI systems, companies can work with educators to fine-tune models based on specific learning goals, classroom contexts, and student needs. This partnership model enables long-term evaluation of how GenAI affects student learning outcomes, particularly in areas such as retention and critical thinking.

An early example of this initiative can be seen in Newark, New Jersey, where the local Board of Education approved "a data-sharing agreement with Khan Academy to study whether [their GenAI-powered tutor, Khanmigo,] was effective 'in the North Ward schools'" (Gomez, 2024). As part of this initiative, researchers analyze state testing data to determine how the use of the companion tool correlates with student growth and academic achievement in grades 5-8 (and later 3-8). The implementation began as a pilot program in 2023 within a subset of schools in Newark's North Ward district, before expanding to dozens of schools and reaching tens of thousands of students—demonstrating a scalable, phased adoption model.

Beyond evaluation, this collaboration also emphasizes how GenAI can be meaningfully integrated into classroom instruction. Particularly, teachers have used Khanmigo to augment their teaching by generating lesson materials, drafting problem sets, and tailoring content to student interests—for instance, creating algebra problems based on familiar cultural references like Pokémon (Khan Academy, 2026). Moving forward, this type of collaboration allows districts to move beyond assumptions about GenAI's effectiveness and instead rely on empirical evidence to guide pro-learning-retention adoption. This also allows tools that have initial focus problems, such as Khanmigo, to cope over time (e.g., from 2024 to 2026) by learning from this same evidence and refining their parameters to align their internal GPT model with the goals of educational institutions.

Constrained AI Design for Active Learning

In addition to external oversight and policy measures, the design of GenAI systems themselves can be modified to better support learning retention. One promising approach is to shift from answer-generating systems to guided Socratic-style tutors that prioritize student engagement over efficiency, as seen recently in emerging implementations from different web platforms. This specific teaching style is associated with **active learning**, an approach “in which all students are asked to engage in the learning process” (University of Minnesota, Twin Cities, 2026).

For instance, Khan Academy’s Khanmigo is intentionally structured to act as a Socratic tutor rather than a solution engine: it guides students through problems step-by-step, asks probing questions, and often refuses to provide direct answers outright. Similarly, tools associated with Duolingo Max, an AI-powered subscription plan from Duolingo, incorporate features such as “Explain My Answer,” which encourages learners to reflect on mistakes instead of passively receiving corrections. These modes demonstrate how AI can be deliberately constrained to reinforce learning behaviors rather than bypass them.

Similarly, across emerging GenAI tools, there is a growing shift toward “guided interaction modes”—including Socratic tutoring (Quizlet Q-Chat), step-by-step reasoning (Photomath), and scaffolded hints (Carnegie Learning). These design choices acknowledge that unrestricted answer generation can undermine learning retention, prompting developers to instead incorporate constraints that encourage active engagement.

In synthesis, as seen in other proposed measures, constrained AI design is already making its way into the tools students and instructors use daily.

Closure

GenAI tools, particularly those offered by EdTech companies, present exciting possibilities for the future of education, but are still far from perfect in supporting long-term knowledge retention. As these systems become more embedded in classrooms, it is critical that their design prioritizes human-centered learning—through guided interaction, teacher involvement, and structured constraints—rather than unrestricted answer generation. Without these safeguards, students risk becoming dependent on AI in ways that compromise their critical thinking, ability to retain information, and engagement in independent problem-solving. These consequences extend beyond the classroom, affecting students’ preparedness for the intellectual demands of higher education and the adaptability required in the modern workforce. Ultimately, the success of GenAI in education will not be determined by how efficiently it delivers answers, but by how effectively it supports sustained learning retention during students’ formative years, and via the collaborative effort of students, families, schools, and GenAI tool providers.

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